

Research Ethics and HCI

Amy Bruckman

What is Research Ethics in HCI?

Consider these ethical challenges of human–computer interaction (HCI) research:

- May I pay homeless people to answer a survey about how they use their cell phones? Is even a small gift card potentially coercive to people with extremely limited means? Is their financial need so great that their ability to make a rational decision is diminished? (LeDantec & Edwards, 2008).
- If I want to study people’s behavior on Facebook, do I have to worry about whether this is allowed by the Facebook Terms of Service? (Gilbert et al., 2008).
- If I am recording data to help make evidence-based decisions in the care of a special-needs child, how do I balance the value of that data with people’s fear of surveillance? (Hayes, 2004; Hayes & Abowd, 2006).

As HCI expands its domain to every corner of the human experience, issues of research ethics keep getting more complicated. HCI has come a long way from the days of laboratory studies where volunteers completed given tasks to receive \$10 gift cards, cash, or course credit. The need for careful attention to ethics in HCI research is growing. Further, research in HCI has expanded its subject and methods to such a degree that aspects of the basic paradigm of “human subjects research” need to be rethought.

In this chapter, I briefly review some core concepts in the ethical treatment of human subjects before focusing on nuances that are important or unique for HCI research. I then highlight challenges raised by research on the Internet.

Research ethics involve understanding complex relationships between law, policy, and ethics. What is codified in institutional policies and procedures is typically a higher standard than merely legal, and what is ethical is a higher standard than

A. Bruckman (✉)

Georgia Institute of Technology, 85 5th. St NW, TSRB 338, Atlanta, GA 30332-0760, USA

e-mail: amy.bruckman@cc.gatech.edu

what is allowed by policy or law. Laws shift at political boundaries, policies shift at institutional boundaries, and ethics shift at cultural boundaries. For this paper, I will focus primarily on research at US universities.

Researchers create three primary risks:

- First, they risk harming their subjects.
- Second, they risk disturbing the environments they are trying to study (as researchers often do; see *Reading and Interpreting Ethnography*, this volume). This may also reduce the group's willingness to cooperate with future researchers.
- Third, they create a risk of serious consequences for their institution if ethical violations occur.

In extreme situations, the US government has suspended all human subjects research at universities for periods of time (Manier, 1999; Weiss, 1999). If one study was not properly regulated, then all studies must be rereviewed since the review process at the institution has been declared suspect. Such extreme actions are rare, but the database of what are called determination letters, descriptions of violations of regulations, kept by the US Department of Health and Human Services Office of Human Research Protection makes interesting reading (OHRP, 2012). The stakes are high for universities.

The History of Human Subjects Research

During World War II, researchers in Nazi Germany used humans in horrific experiments that today we would not allow to be performed on animals. In response, the Nuremberg Code was drafted in the wake of The Nuremberg War Trials (1949), the trials that revealed these atrocities. This established the basic principles that subjects should provide voluntary consent, and be free to end their participation at any time. Further, the experiment must “yield fruitful results for the benefit of society,” and unnecessary risk should be avoided.

However, problems still existed in the ethics of human subjects research. Most notably, in the Tuskegee syphilis experiments, 400 African-American men with syphilis were monitored for 40 years (1932–1972) without being told of their disease (UNLV, 2012). The study continued for decades after the discovery in 1947 that penicillin cures the disease (CDC, 2011). When this atrocity came to light, a commission was created which led to the creation of The Belmont Report (Belmont, 1979). The Belmont Report remains the primary statement of principles of ethical research on human subjects today. It lays out three main principles:

1. Respect for persons

- (a) Treat people as ends in themselves, not means to an end.

2. Beneficence

- (a) Do not harm, and
- (b) Maximize possible benefits and minimize possible harms.

3. Justice

- (a) Distribute the burdens and benefits of research equally across society.

The first principle draws on Immanuel Kant's categorical imperative, which says that you must "act in such a way that you treat humanity, whether in your own person or in the person of any other, never merely as a means to an end, but always at the same time as an end" (Kant, 1964). For Kant, this is the essence of what it means to be moral. In the context of research ethics, this means seeing research participants not as a means to obtain data but as morally relevant beings whose autonomy and needs matter.

The second principle is somewhat misleading in its wording, because "do not harm" is too absolute. There are risks in simply walking across a room—you might trip and fall. But if a subject trips and falls (due to no special risks created by the study), that does not mean the study was unethical. People can be harmed through their participation in ethical studies. The challenge for evaluating the ethics of a proposed study is to weigh possible benefits and harms. An Institutional Review Board (IRB) member once told me the story of a study his IRB rejected that had no risks beyond those of participating in every day life. The board judged that the research design was so weak that it had no potential benefits to research. In this case the miniscule risk outweighed the nonexistent benefit and the study proposal was rejected. The ethics of a study and the research value of a study are inseparable concepts.

Weighing the benefits and risks of a study is made more challenging by the fact that disparate things are being compared, and the benefits and costs typically fall on different individuals. The third Belmont principle, justice, asks us to try to address this problem by distributing benefits and risk equally across society. In practice, the principle of justice is the most neglected of the three in a variety of ways. For example, people of low economic means are the more likely to agree to participate in risky studies in return for monetary compensation. Pregnant women are often excluded from studies for fear of added risk to them or their fetus, sometimes in cases where no such added risk exists. Children, defined as people below the age of 18 years of age, may be excluded simply because obtaining parental consent creates extra work for investigators. To begin to address these challenges, some human subjects' review boards now require explicit justification when groups such as pregnant women and children are excluded from a study population.

In the USA, US code Title 46 contains the laws regulating federally funded human subjects research (OHRP, 2001). Most universities in the USA apply these regulations to all research they conduct, even though technically the rules apply only to federally funded work. US corporations are subject to these laws only if they accept federal research funding. Some corporations have their own human subjects review boards, and some do not. Corporate review boards are not required by law and do not need to meet government standards.

Universities in the USA that accept federal funding are required to maintain an IRB committee. IRB members must represent diverse disciplines, and must include a non-scientist and a member of the community not affiliated with the university (US §46.107). Members typically take on a relatively heavy service load, and are required to understand issues in a wide range of disciplines, predominantly outside their own areas of expertise. Investigators seeking to work with human subjects must first obtain certification in human subjects research, typically by taking a short course (which is often offered online). The Collaborative Institutional Training Initiative (CITI) is one organization that offers training (see <http://www.citiprogram.org>). Next, they submit a proposal to do human subjects research to their institution's IRB for review. Some IRBs will as a matter of policy request at least one change to every protocol submitted, to clearly document that the protocol was actually reviewed and not simply approved without scrutiny. While the primary job of an IRB is to protect human subjects, a secondary and legitimate function is to protect their institution from liability and official sanctions.

The Evolving IRB

IRBs were originally created with medical and psychological research in mind. Policies and procedures developed in this context do not always translate easily into new research domains, like HCI. Weighing the benefits and risks of a study is impossible without understanding the domain. In the 1980s and 1990s, many IRBs had limited knowledge of computing and the Internet. IRBs are made up of people with different backgrounds and knowledge, and each board over time becomes comfortable with certain kinds of research they have seen repeatedly. An institution that rarely reviews HCI research may have more difficulty reviewing such protocols than one (like our IRB at Georgia Tech) that reviews them on a regular basis. Each IRB has more experience and comfort in some areas than others. In other words, an IRB is a kind of knowledge-building community of practice. HCI researchers versed in theories of sociotechnical systems will find their knowledge is useful in helping them to understand their IRB.

As computers have become pervasive, most IRBs have developed experience and knowledge of HCI research. If your research area is somewhat unfamiliar to your IRB, then you can request to make a short presentation to the board to help them become better informed. If your IRB seems to be erecting unreasonable barriers to HCI research, your best recourse is to *join them*—to volunteer to be an IRB member, and change the culture from within. Numerous faculty have taken this step, for example Richard Nisbett at University of Michigan and Robert Kraut at Carnegie Mellon University. They were able to greatly advance HCI research at their universities while maintaining high ethical standards (Kraut, 2012; Nisbett, 2012). Joining the IRB is more advisable for a tenured faculty member, because political complications may occur.

You may also have the opportunity to place a thoughtful graduate student on the IRB as a student representative. This is a fantastic learning opportunity for the student, and a series of students from the College of Computing have made strong contributions to Georgia Tech's IRB over the last 15 years.

Nisbett and others later participated in a board organized by the National Research Council (NRC) that produced a draft report calling for changes in our protections for human subjects. Conversations begun by this report are ongoing. The process of official change is slow and requires extensive effort on the part of concerned stakeholders.

IRB members accept a huge service burden on behalf of the community, and struggle to keep up with the broad areas of knowledge they need to understand. They accept significant legal liability by assuming this post, and are typically offered special insurance coverage by the university¹ (Bordas, 1984). It is a challenging and often thankless job, and they generally appreciate anything that makes it easier for them to do their work well.

In approaching an IRB, an attitude of humility and sincere concern is essential. If you sincerely care about ethics and see the IRB as partners in collaboratively solving a difficult problem, they are more likely to be receptive to your proposals. Many IRB members will make themselves available to discuss nuances of a proposal before it is submitted. The IRB process is a ideally collaborative and mutually supportive among board members and applicants.

For HCI professionals considering employment at a particular university, asking questions about the institution's IRB is strategic. With patience and education, well thought-out proposals can win approval by almost any IRB. However, what is quick and easy at one institution may be challenging and time consuming at another.

Balancing benefits and costs of research can be particularly challenging in the context of emerging technologies. In 2001, MIT Media Lab researcher Pascal Chesnais requested IRB permission to do research on e-mail filtering. Though it is hard to comprehend this now, at the time unsolicited e-mail was relatively rare. Chesnais's work would have been an early contribution to the new field of spam filtering. Unfortunately, the MIT IRB denied Chesnais permission to do the research. The IRB members determined that if an individual uses Chesnais' e-mail filter, it affects whether he or she receives all mail sent. Clearly the study volunteers could consider whether this risk was acceptable and make an informed decision on whether to participate. However, the board also reasoned that this affects *everyone who could potentially send the volunteer e-mail*. If I try to mail you and my message does not arrive, then I have been negatively affected by the study; however, I did not choose to opt into the study. Therefore, the board reasoned that everyone who could potentially send a user mail also would need to give their consent. After more than a year of arguing with the IRB, Chesnais abandoned his study.

At this time, the technology was new, and Chesnais was ahead of his time. His proposed study had both potential risks and potential benefits. As we have seen,

¹IRB members employed by state institutions may be protected by the doctrine of sovereign immunity. Exact laws vary by state. See Bordas (1984) for a detailed discussion.

research ethics is about balancing risk versus benefit. The need for e-mail filtering was not yet well understood, and hence the risk–benefit scale tipped towards unjustified risk. Anyone who has ever failed to receive an important e-mail message because of misbehavior of a spam filter understands that risk is real. The nature of the potential benefit only became clear over years that followed as we all became inundated with advertisements for pornography, malware, and scams. The IRB’s decision was unfortunate but not unfounded in its reasoning. Chesnais needed both to acknowledge the risks and make a more compelling argument about the coming epidemic of unsolicited e-mail and potential benefit of his research.

Process

The first step for any researcher is to determine whether you are in fact doing human subjects research. According to the US code (§46.102), a human subject is “a living individual about whom an investigator (whether professional or student) conducting research obtains (1) data through intervention or interaction with the individual, or (2) identifiable private information.” Research means “a systematic investigation, including research development, testing and evaluation, designed to develop or contribute to generalizable knowledge.” (See <http://www.hhs.gov/ohrp/human-subjects/guidance/45cfr46.html>). Under these definitions, for example, simply analyzing publicly available information on the Internet is not human subjects research. If you are not doing human subjects research, you are not formally held to human subjects procedures and regulations; however, you still should take care to proceed in an ethical fashion.

If you are doing human subjects research, the normal process for obtaining permission to do human subjects research at a US university starts with the researchers obtaining a certification in human subjects research, obtained typically by reading material online and taking short quizzes to ensure understanding. Next a proposal to do research is submitted, to be reviewed by the university’s IRB.

An IRB proposal includes a description of the research and consent forms that the subjects will read and sign. Consent forms must explain any risks of participating, any benefits, a statement that subjects can withdraw at any time, and more. It is important to check your IRB’s exact requirements. The purpose of the form is to help potential research volunteers make an informed decision about whether to participate.

IRBs generally have a regular meeting schedule, and deadlines for when materials need to be received to be considered at each meeting. Checking the calendar is important so you do not end up accidentally missing the deadline and having to wait an extended period of time for your protocol to be reviewed.

There are three levels of IRB review: exempt, expedited, and full board. When you file an IRB proposal, you typically may request which kind of review you believe is appropriate; however, your IRB will make the final determination. For research at US universities, exempt categories are defined by law in

§45CFR 46.101(b)(1)-(6). They include research “conducted in established or commonly accepted educational settings, involving normal educational practices,” research on existing public records or where individuals are unidentifiable, and others. If what you are doing is human subjects research, you still need to file an IRB application and your IRB determines whether your proposed research is exempt. Investigators are not allowed to determine this for themselves. However, in practice, it is often a fine line to judge between research which should be proposed and declared exempt versus work that is not even technically human subjects research. The formal definition of human subjects research discussed above should be carefully applied to any research to help make this determination.

If a proposal is not exempt, it may qualify for *expedited* review. Expedited review means an IRB staff person may quickly approve the protocol without sending it to the full board. This saves significant time, because full board meetings typically take place only once a month (or even less often) and a protocol that just misses the deadline may wait longer. Much HCI research qualifies for expedited review. Work that qualifies for expedited review is defined in §46.110 (see <http://www.hhs.gov/ohrp/policy/expedited98.html>). Expedited review includes “Research on individual or group characteristics or behavior (including, but not limited to, research on perception, cognition, motivation, identity, language, communication, cultural beliefs or practices, and social behavior) or research employing survey, interview, oral history, focus group, program evaluation, human factors evaluation, or quality assurance methodologies.”

In designing an IRB protocol, an important factor to consider is whether vulnerable persons might be involved. Children and individuals with “diminished autonomy” (including those with cognitive disabilities) are not permitted to give consent, since they are not allowed to enter into legal contracts. Instead, they may be asked to assent, and are given an assent form written at a level appropriate for their reading level. Their parent or guardian must sign a consent form on their behalf. Checking reading level is advisable for both consent and assent forms; many word processors provide tools to evaluate the reading level of a document. A common rule of thumb is that documents should have an eighth grade (age 14) reading level or lower (See http://en.wikipedia.org/wiki/Literacy_in_the_United_States).

To the strictest standard, a consent form must be signed by the volunteer, the subject, and a witness. The lower the risk of the study, the more these requirements may be relaxed. In many low risk studies, an IRB may waive the requirement to document consent. US §46.117 states:

- (c) An IRB may waive the requirement for the investigator to obtain a signed consent form for some or all subjects if it finds either:
 - (1) That the only record linking the subject and the research would be the consent document and the principal risk would be potential harm resulting from a breach of confidentiality. Each subject will be asked whether the subject wants documentation linking the subject with the research, and the subject’s wishes will govern; or
 - (2) That the research presents no more than minimal risk of harm to subjects and involves no procedures for which written consent is normally required outside of the research context.

Studies which allow for a “click to accept” Web-based consent forms require a waiver of documentation of consent, since clicking on a Web form is not legal consent. Note that the above refers to waiving the requirement for *documentation* of consent. Waiving the requirement for consent entirely is more rare, but may be permitted if the study is low risk and could not be carried out without the waiver (US §46.116). For example, I obtained a waiver of consent to conduct a qualitative study of children’s learning in an after-school computer clubhouse. Different children attend the drop-in program each day. For months we sent members home with forms for their parents to sign; however, few were returned. We eventually applied for and were granted the waiver. (The study was also arguably exempt from IRB review as a “normal educational setting” under §46.101).

It is important to keep careful research records. IRBs may audit a study to verify that it is in compliance with the approved research protocol.

The Nature of the Subject in HCI Research

The traditional paradigm of human subjects research presumes an expert investigator interacting with subjects who are lay people with no special knowledge of the research domain. The subjects are viewed as vulnerable individuals who will be compensated appropriately for their contribution to research, and anonymized in written accounts of that research for their protection. These assumptions are increasingly anachronistic. The nature of any vulnerability depends on the nature of the research, and many kinds of HCI research were simply not envisioned when the rules were first written.

In many research areas, our subjects are increasingly knowledgeable about our domains of inquiry. To give an easy example, if we are developing software to support a more efficient work process, then we might employ a participatory design method (Ehn, 2008), working with engineers at our study site as design partners. In this case, the professionals who helped with the design may wish to comment on drafts of any publications that result. Depending on the exact circumstances, they may prefer to be credited for their work rather than anonymized in written accounts. They may even want to be co-authors (See [Knowing by Doing: Action Research as an Approach to HCI](#), this volume). If they have significant input, to deidentify them is to deny them credit for their work. As a result, in some circumstances, anonymizing can *cause* harm rather than protect individuals from harm.

Participatory design in partnership with professional engineers seems like a far cry from anthropologists studying a tribal culture or medical researchers trying out a new antibiotic or medical device. However, the situations are not as different as they may appear. The medical literature increasingly views patients as partners who must accept responsibility in managing their own care (Mamykina & Mynatt, 2007). Patients now often may read obsessively about their own conditions, and are increasingly encouraged to not simply blindly accept instructions from healthcare providers. The Internet has brought them a wealth of information (and misinformation),

and current thinking in public health suggests that patients fare better when they are not passive in managing their care. If patients are collaborators in managing their own care, participants in studies of new medical systems in a sense become collaborators in that research. Our view of who “ordinary people” are and what their role in the research process might be has evolved.

Our fundamental view of the nature of “the research subject” has shifted significantly. A post-modern interpretation of research practice situates these shifts in a broader intellectual framework. For example, the anthropologist Renato Rosaldo writes:

The Lone Ethnographer’s guiding fiction of cultural compartments has crumbled. So-called natives do not “inhabit” a world fully separate from the one ethnographers “live in.” Few people simply remain in their place these days. When people play “ethnographers and natives,” it is ever more difficult to predict who will put on the loincloth and who will pick up the pencil and paper. More people are doing both, and more so-called natives are among ethnographer’s readers, at times appreciative and at times vocally critical. (Rosaldo, 1993, p. 45)

A critical view of the subject brings the Belmont Report’s first principle of “respect for persons” into new focus. The notion of ignorant subjects incapable of understanding or responding to our representations has always been misguided to a degree, but is even more so in the age of the Internet. For subjects who are literate and computer literate, our scholarly works are increasingly accessible, and our subjects are more and more likely to be both able to and interested in engaging with our research findings.

As a result, it is important for the ethical researcher to rethink power relations between researchers and participants, and offer participants more voice (Borning & Muller, 2012). Subjects may wish to be named in written accounts. This is particularly true, for example, in work my students and I have done on online content creators, who are justifiably proud of their creations. As a result, most of our consent forms contain this statement:

In some cases, people we interview are proud of things they have done online (for example creative projects) and would like to have their real name listed in our published reports. If you would like to request that we use your real name if possible, please sign below. We will not be able to use your real name if we feel there is anything that might embarrass you in our report. For most people, using a fake name (that we invent) is the right choice, so you do not need to sign here but you may if you wish.

If subjects do nothing, we anonymize them as usual. They must sign to have us use their real name, and the word “optional” appears next to the signature line in case people are quickly signing without reading the content.

There are some nuances to the above wording. Respect for persons suggests that we honor the wishes of our subjects and name them if they wish to be named. However, people do not always make good choices about when to seek publicity, as we see for example in individuals who allow themselves to be filmed being arrested on reality television shows. Individuals sometimes put an undue value on public recognition and make unwise decisions about what to publicize about themselves. In this case, the value of “respect for persons” conflicts with the value to minimize

harm. To suggest that we rethink knowledge and power differentials between researchers and subjects does not mean that those differences do not exist—sometimes researchers really do still know better. The question of legal liability also comes into play. Even if the individual requested their name be used, the university might be liable if the portrayal of a subject caused harm. As a result, whether to publish real names when requested is ultimately a judgment call that rests with the researcher.

If our subjects are knowledgeable individuals who understand the area of study, they may be capable of and interested in responding to drafts of research results before publication. Since they may obtain copies of research results and express opinions about those results after publication anyway, it is in everyone's interests to solicit that input before publication when it still might be taken into account. This is more relevant for qualitative studies with descriptive accounts of individuals and groups rather than simply aggregate statistics. Subjects often will not follow through on sending comments in a timely manner, so it is important to give subjects a reasonable amount of time and clear notice of when that time will be up.

In practice, this process is often unproblematic and improves the quality of the research. Subjects may often be able to correct researcher misconceptions. However, in some cases this process becomes quite complicated. Individuals may resist interpretations that they find unflattering. Resistance leaves the researcher with two dilemmas. The first dilemma is epistemological: how do I know who is right? Is the subject just being defensive, or did I really misinterpret the situation? There may not be one correct answer. As Rosaldo notes, listening to many voices often means they can not be seamlessly integrated into a coherent, single, master narrative (Rosaldo, 1993). The researcher has dual ethical obligation to produce accurate research results, and to minimize harm. Ideally obtaining more data can help resolve any misalignments. Alternatively, the existence of multiple, irreconcilable views can simply be acknowledged.

The second part of the dilemma is pragmatic: Do I want to work with these subjects again in the future? Will this account endanger an important relationship? It is unethical to publish an inaccurate account, but strategic omissions can sometimes help resolve such dilemmas. If a controversial or unflattering piece of information is not strictly relevant to the research question being addressed, then it can be safely omitted. For example, in one early draft of a paper about an online system, I described a regular user as “socially awkward” in real life. While that comment provides context to understand the person and his participation in the online community, it was not essential to our research questions about use of the system and was ultimately omitted.

Ethical Challenges of Internet Research

Doing research on the Internet introduces a host of surprisingly complicated and intriguing ethical challenges (Ess, 2002; Ess & T. A. E. W. Committee, 2002; Berry, 2004; Kraut et al., 2004). The first challenge is to even determine when you are in

fact doing “human subjects research.” Accessing information posted on the Internet accessible without a password is not technically human subjects research according to IRB definitions. That information is published, and may be freely used for any purpose without IRB review. US §46.102 says:

- “(f) *Human subject* means a living individual about whom an investigator (whether professional or student) conducting research obtains
- (1) Data through intervention or interaction with the individual, or (2) Identifiable private information.”

If information is published online, you are not interacting with its author and the information is not private. However, should you decide to ask the author a question about their online posting, then you have interacted and are indeed engaged in human subjects research.

This situation is unfortunate, because it *creates a disincentive to conduct high-quality research*. The quality of research is inevitably improved by asking questions of content creators. However, doing so requires significant extra work on the part of the investigator such as writing a human subjects research protocol, waiting for approval, and obtaining consent from subjects. It is not surprising that many investigators choose not to ask any questions, and write maddeningly incomplete papers with statements like “We speculate that users might engage in this behavior pattern because....” Speculation is in fact not necessary, because the users are often available to comment if one only takes the time to get approval to ask them.

Once one does start interacting with Internet content creators as human subjects, another complication emerges: their online postings may be findable with a search engine. This creates a linkage between confidential human subjects research data and their real identities. This is not a problem if the subjects wish to be identified by name. However, if they wish to be anonymized, then we have a dilemma. For studies involving controversial subject matter and significant risk to the participants, the solution is to not quote any online postings of the participants.

One clever trick is to present the same individual as two separate people in the written account—the quote is included but not connected to the person interviewed. Alternatively, if the subject matter is uncontroversial, then a kind of “light disguise” is possible, where a determined person could unmask the subject’s identity but it is not immediately obvious without some effort (Bruckman, 2002). Research protocols from my lab warn of this possibility:

To protect your confidentiality, your name will not appear in any publications; a pseudonym (a fake name) will be used instead. However, in the case of quotes from things you have done online (such as blog entries, forum posts, etc.), this disguise could be vulnerable—a determined and skillful person could potentially break it. Since many online sites are open to the world, search engines (i.e. Google) index them. As such, a person could take a quote and use a search engine to find the actual page, thereby breaking the pseudonym disguise. We do not anticipate that this research will uncover sensitive information, but in case it does, we will omit direct quotes that could be found in a search engine.

Disguising subjects in written accounts often requires omitting or changing identifying details and names. In some studies it is possible to omit the name of the online site being used; however, this is impractical in many cases. For very large

sites or sites with few competitors, the site may be obvious. It is also not possible to disguise the site in design-based studies where researchers are studying an online service they created. Disguising subjects is about striking an appropriate balance. Where disguising is necessary, the quality of the research is improved by disguising as little as possible; however, the privacy of the participants is maximized with more disguising.

Terms of Service

So far I have discussed information available online without a password. If it is necessary to log in to a service to access data, then you are arguably subject to the site's Terms of Service (TOS). The TOS may set conditions for when research is allowed, or may prohibit research altogether.

Abiding by the TOS is always a safe choice. However, some researchers argue that it is not always necessary or appropriate. First, some clauses of the TOS may not be legally valid or reasonable. Companies often include clauses in legal disclaimers that would not hold up in a court of law, and also would not hold up to ethical scrutiny. For example, changes that Google made to its privacy policy in 2012 are being disputed as not complying with European Union (EU) privacy law (<http://www.guardian.co.uk/technology/2012/oct/15/google-privacy-policy>). The terms of service state that "By using our Services, you agree that Google can use such data in accordance with our privacy policies." (<http://www.google.com/intl/en/policies/terms/>) What is the status of that agreement for EU citizens if the privacy policy is not in compliance with EU law?

More importantly, the independence and intellectual integrity of the research community is at stake. Imagine that a large Internet site that is reshaping business and culture decides to erect major barriers to research. Does the site's regulations mean this aspect of the human experience is off limits for scholarly inquiry? Further, a corporation might allow researchers access only on their own terms, and permit work (or its publication) only if it is flattering to the company. Internet researchers are increasingly dependent on the good will of corporations in providing access to data, and the intellectual independence of the academy is challenged by this state of affairs.

So if a site's TOS may not be valid and you do not accept the corporation's authority to control what is said about it, is it acceptable to ignore an Internet site's TOS? Doing so may have practical consequences (you may be denied further access) or legal ones. My cautious answer is that it may sometimes be permissible to ignore TOS, but I would only do so after careful reflection (and with sound legal advice). The rationale for doing so should also be explained in the publication.

The unresolved quandary for the research community concerns how reviewers of scholarly manuscripts should handle this issue. In some cases, work has been rejected from conferences and journals for failing to comply with TOS. The same

work with different reviewers might be accepted. This inconsistency is unfortunate. The situation would be improved if conference committees and professional societies establish clear standards and apply them consistently.

More generally, reviewers often struggle with whether the ethics of a study are within their purview. At one program committee meeting I attended, a program committee member argued that the study in question had passed IRB review at the author's institution, and that was sufficient. After all, the researchers got approval that their approach was ethical and proceeded accordingly. This heuristic would greatly simplify the task before reviewers; however, it is built on the assumption that local ethics review is being done well everywhere. If reviewers are expected to double check ethical issues in research, then they need clear standards to guide them. Ethical codes by organizations like the ACM (ACM, 1992) and IEEE (IEEE, 2012) are candidates, but can sometimes be too general to help with specific challenges, and also may not be updated frequently enough to account for emerging issues. The role of program committees in double-checking ethics has not been clearly resolved in many research communities.

Recruiting Subjects Online

Recruiting subjects and sampling data for any study are critical processes that affect the quality of research results. Internet-based recruiting introduces the additional challenge that the researcher may disturb the research environment through the process of soliciting participants. In the early days of scholarly interest in users on the Internet, members on some sites joked that the researchers might outnumber the natives. In the Georgia Tech course CS 6470 Design of Online Communities, students do participant observation and interviews with members of an online site. Although we do our best to teach students how not to annoy site members through the process of requesting interviews, students are still occasionally asked to leave a site because their requests for interviews are perceived as disruptive. Requests for research participation both impose on people's time and also foster a self-awareness that may disturb normal patterns of online behavior. Students in the online communities class are given a general rule of thumb is that any online participant should see a request for study participation only once or at most twice. As a result, targeted inquiries sent to specific individuals are often more effective than broadcast messages seen by large groups.

Researcher presence and request for study participation is more likely to be perceived as disruptive if the researchers are seen as not belonging on the site. For example, two student researchers were not welcomed by a divorce support forum they attempted to study because the students were themselves happily married. Although all their communications were seemingly appropriate in tone and content, their lack of personal experience with divorce made them unwelcome. This is more likely to occur in topics that address sensitive subjects like divorce or health support. For an extended discussion of this topic, see (Bruckman 2012).

In recruiting subjects online, researchers need to be aware that responders could be anyone and anywhere. Is the volunteer a child? Do they speak fluent English and fully understand what they are volunteering for? It is important to explicitly ask people's age. IRB protocols that allow a waiver of documentation of consent with a "click to accept" Web consent form for adults often typically do not allow this procedure for minors. In our protocols, we use paper parental consent and child assent forms that must be signed and scanned or sent to us by fax.

A growing number of studies are relying on crowd sourced participants such as workers on Amazon Mechanical Turk. On such sites, people pay a small amount of money (often a few cents) for workers to complete simple tasks. Workers often receive compensation less than the legal minimum wage in the USA, and are sometimes denied compensation for work legitimately completed. The invisibility of such workers is arguably antithetical to the basic principle of respect for persons (Silberman, 2010; Bederson & Quinn, 2011). These factors need to be taken into consideration when using crowd sourced labor either as a source of research subjects or for assistance in processing research data.

Recording Ephemeral Interaction

The discussion so far has concerned content that is archived online. The issues change (and become more controversial) when we consider researchers recording otherwise ephemeral communications. Some research communities (particularly linguists) argue that it is ethical to record conversation of non-identified individuals in public places. By analogy, chat rooms and other ephemeral forms of communication may be viewed as a similar situation, particularly if user names are removed from the data (Herring, 1996). However, such recordings often make Internet users angry, violating their expectations of privacy (Hudson & Bruckman, 2004). Privacy depends on the idea of "reasonable expectations," but with new technologies expectations are emergent, may not be shared between researchers and the general public, and are evolving.

Is it ethical to record activity in a chat room or a virtual world without consent? Does this constitute activity of unidentified individuals in a public place? One compromise solution suggests that traces of online interaction may not be *recorded* without permission; however, a participant observer may take field notes on his/her own experiences without permission. Others encountered would then be anonymized in written accounts. This compromise is splitting hairs because as soon as the participant observer asks explicit questions about the environment for research purposes, then informed consent is required. There is a thin boundary between normal conversation as a participant observer and interviewing.

In some cases, the process of requesting consent for recording may be more disruptive to the online environment than the study itself. This is particularly true in synchronous communication media. For example, suppose you want to study a synchronous chatroom. Can you ask participants to either opt in or opt out of the study

by typing a command? In our chatroom study (Hudson & Bruckman, 2004), we addressed this question and found that few people opted in, few opted out, and in fact the researcher was kicked out of the chatroom in most cases because the process of requesting participation annoyed members. Researcher presence was perceived as less intrusive in larger groups. For every additional 13 people present, the chance of being kicked out dropped by 50 %. In cases in which the process of requesting participation is itself disruptive, if a study is low risk, researchers may have a compelling case for a waiver of consent.

In 2012, the preponderance of online interaction is archived, in either synchronous or semi-synchronous modes. A relatively small number of researchers study real-time environments such as Massively Multiplayer Online games (MMOs) where researchers may desire to record otherwise ephemeral communication. However, going forward, the increase in data usage from mobile tracking systems like GPSs and increasingly powerful RFID technology will likely raise ethical challenges. For example, location-based data can play a role in a variety of legal cases (especially divorce lawsuits), and the creation and storage of that data creates real liability and risk. Invisibility of data collection can foster a gap between how much privacy citizens expect and how much privacy some experts judge they are entitled to expect. Just as we can ask whether someone using a pseudonym in a chatroom is really an unidentified person in a public place, we can ask the same question of an individual carrying an RFID tag in a mall. The challenges are similar and answers are not entirely clear.

Design Ethics

HCI research involves creating new artifacts that reshape the human experience. Shaowen and Jeffrey Bardzell quote Papanek, commenting that “Design is intervention, an intentional effort to create change. As design theorist Papanek defines it, design’s job is ‘to transform man’s environment and tools and, by extension, man himself.’” The Bardzells note that “HCI is increasingly engaging with matters of social change that go beyond the immediate qualities of interaction. In doing so, HCI takes on scientific and moral concerns” (Bardzell & Bardzell, 2011). Batya Friedman proposes a model of Value-Sensitive Design (VSD) where examining ones values and the values inherent in the design is an explicit step in the design process (Friedman, 1996; Friedman & Kahn, 2006; Borning & Muller, 2012).

In truth, all design has embedded values. For example, consider something as seemingly innocuous as a word processor. Jeanette Hofmann details how the design of word processing software has embedded conceptions not only of the level of technical sophistication of the user, but what it means to be a writer and who is expected to write. Early word processors assumed a secretary was typing text written by an executive. A system like WangWriter assumed that secretary had limited ability to learn commands. The resultant system made it unlikely to make a mistake but also impossible to do anything quickly. In contrast, systems like WordPerfect

and WordStar assumed that the secretary was a professional who had the time and intelligence to learn complex commands, leading to greater productivity and a much wider range of capabilities after a significant initial learning curve. It was not until the creation of the word processor on the Xerox Star that the concept emerged of a knowledge worker who would work directly on a keyboard. And paradoxically, the Xerox Star assumed the knowledge worker was a computer dilettante who had limited time to learn commands. Assumptions about the capabilities and goals of typists and writers shaped the tools which in turn significantly shaped the lives of those using the tools (Hofmann 1999). Design ethics challenges researchers to answer the question, what kind of world are we creating?

A full discussion of this topic is beyond the scope of this chapter, but see [Knowing by Doing: Action Research as an Approach to HCI](#), this volume, for a key perspective. This topic is of growing importance—it would be difficult to overstate its significance.

Conclusion

Ethical research on human subjects in any field involves balancing the potential benefits with potential harms. New technologies can make it more challenging to determine the nature of the potential benefit (like the need for a spam filter) and the nature of the potential harm (like violating someone's privacy). Traditional human subjects research seeks to protect vulnerable research participants by anonymizing them in written accounts. However, in a growing number of circumstances, our participants may want credit for their work. To anonymize them would be to cause harm. Further, our participants are increasingly qualified to and interested in responding to our representations of them.

Simply analyzing data available on the Internet without a password does not technically constitute human subjects research. However, it often substantially improves the quality of the research findings to interview or survey or otherwise interact with creators of online content, and this *does* constitute human subjects research. It is inappropriate to speculate about what users were intending without taking the time to ask them.

A number of issues in HCI research on human subjects still pose unresolved challenges. Is it OK to sometimes ignore Terms of Service of an online site? How do we maintain the independence of the academy from corporate control of access to research data? Is it OK to record unidentified activity in a public place even if that makes people angry if they know this is taking place? Should research reviewers double-check the ethics of a study, or is local ethical review sufficient? If reviewers should check the ethics of submitted work, by what standard? Since legal and ethical standards vary across countries and even among institutions within a single country, whose standards should be applied? Leadership is needed in resolving these issues from both individuals and professional societies. And emerging technologies will inevitably create new challenges that have not yet been named.

Research ethics for HCI has evolved in a tremendously positive way over time. Regulators have become better informed about computers and HCI, and ethical norms have evolved to catch up with challenges created by new technologies. At the same time, HCI research has pervaded every day life in new ways that make the need for regulation more apparent and accepted by the research community. Finally, in the dynamic context of HCI, our basic concept of what it means to do “human subjects research” has evolved in ways that have implications for other fields.

Useful References

A good general review article on research ethics in psychological studies came out of an American Psychological Association (APA) workshop led by Robert Kraut (Kraut et al., 2004). Many qualitative methods texts have thoughtful commentaries on research ethics. I recommend the ethics chapter in *Interviewing as Qualitative Research* by Irving Seidman (Seidman, 2006).

Some of the most impressive positive examples of research ethics done well concern people successfully working with vulnerable populations in a thoughtful and respectful manner. For example, Christopher LeDantec studied use of communications technologies by the homeless, and did a noteworthy job of understanding the power relationships between homeless and researchers and accounting for them in his study design (LeDantec & Edwards, 2008). Similarly, Gillian Hayes has done impressive work in understanding the needs of autistic children and their families. Her intervention uses recording technologies to help caregivers understand the cause of behavioral outbursts, and significant effort went into careful management of the privacy implications. For recording data in school settings, other students and teachers who are not participating in the study may inadvertently appear in the recordings. In general, people’s fear of surveillance may interfere with the significant benefit of having data to make evidence-based care decisions. Careful rules about data collection, retention, and use is needed to address these challenges (Hayes et al., 2004; Hayes & Abowd, 2006). These are excellent models for how to handle tricky situations.

While those are strong examples of situations where extra care is needed, at the other end of the spectrum researchers might consider that sometimes more is possible than they might think. For example, Jim Hudson and I were granted a complete waiver of consent for a deceptive study with no debriefing for our study of how chatroom participants feel about being studied. We most definitely annoyed people with our study. However, our IRB was persuaded that the benefits of the study outweighed the burden on users (Hudson & Bruckman, 2004).

Finally, one good example of how to conduct research on sites with restrictive Terms of Service is work by Eric Gilbert comparing the social networks of rural versus urban Facebook users. To comply with Facebook’s terms of service, Gilbert had subjects come to the lab to log in and show their information to researchers. It is against the Facebook TOS to scrape information from Facebook, but if the users log in they voluntarily reveal information (Gilbert et al., 2008).

My Involvement in Research Ethics

I first got interested in questions of research ethics because I had to. My PhD dissertation, MOOSE Crossing, was a text-based virtual world for kids. When I started to research how people usually handle research ethics in such situations, I discovered there were no precedents. What I was doing was new. I had to collaborate with my IRB to invent a reasonable approach.

My institution's IRB at that time typically erred on the side of caution and worked from a rather expansive definition of reasonable expectations of privacy. I remember being requested to present my study to the full IRB board meeting. I was terrified. A turning point in the conversation was when I explained to the board that I would not be giving Internet access to any children. The participants had to already have Internet access, and I was providing a (somewhat) safer activity for them to do online.

That initial conversation took place in 1993. Twenty years later (at the time of this writing in 2013), I find we are still struggling with these issues. My current institution recently required me to take an IRB refresher course. The online course contained a new module on Internet research, and the content mostly said things were not yet clear to anyone! In the face of these uncertainties, researchers must understand the core ethical principles and construct a well-reasoned argument about how to apply them in any given situation.

Exercises

1. What is the difference between exempt research and expedited research, according to IRB rules?
2. What populations of people would you be studying if the IRB review has been neither exempt nor expedited?
3. In your institution, what is the minimum turn-around time for getting feedback on an IRB application? How often does the committee meet? Is there a separate board that reviews medical from behavioral applications? Is there a separate form? What kinds of questions are asked on a medical application that are not relevant to a behavioral application?

References

- (1949). *Trials of war criminals before the nuremberg military tribunals under Control Council Law No. 10*. Retrieved June 13, 2012 from <http://www.hhs.gov/ohrp/archive/nurcode.html>
- ACM (1992). *ACM code of ethics and professional conduct*. Retrieved June 13, 2012 from <http://www.acm.org/about/code-of-ethics>
- Bardzell, S. & Bardzell J. (2011). Towards a feminist HCI methodology: Social science, feminism, and HCI. *Proceedings of CHI* (pp. 675–684). Vancouver, BC, Canada: ACM

- Bederson, B. & Quinn A. (2011). Web workers unite! addressing challenges of online laborers. *Proceedings of alt.CHI*. Vancouver, BC, Canada: ACM
- Belmont. (1979). *The belmont report*. Washington, DC: US Department of Health, Education, and Welfare.
- Berry, D. M. (2004). Internet research: Privacy, ethics and alienation—An open source approach. *Internet Research*, 14(4), 323–332.
- Bordas, L. (1984). Tort Liability of Institutional Review Boards. *West Virginia Law Review*, 87(1), 137–164.
- Borning, A. & Muller M. (2012). Next steps for value sensitive design. *Proceedings of CHI* (pp. 1125–1134). Austin, TX: ACM
- Bruckman, A. (2002). Studying the amateur artist: A perspective on disguising data collected in human subjects research on the internet. *Ethics and Information Technology*, 4(3), 217–231.
- Bruckman, A. (2012). Interviewing members of online communities: A practical guide to recruiting participants. In A. Hollingshead & M. S. Poole (Eds.), *Research methods for studying groups and teams* (pp. 199–210). New York: Routledge.
- CDC (2011). *The Tuskegee Timeline*. Retrieved October 31, 2012 from <http://www.cdc.gov/tuskegee/timeline.htm>
- Ehn, P. (2008). Participation in design things. *Participatory Design Conference* (pp. 92–101). Indianapolis, IN: ACM
- Ess, C. (2002). Internet research ethics. *Ethics and Information Technology*, 4(3), 177–188.
- Ess, C. & T. A. E. W. Committee (2002). *Ethical decision-making and internet research*, *Association of Internet Researchers*. 33
- Friedman, B. (1996). Value-Sensitive Design. *ACM Interactions*, 3, 17–23.
- Friedman, B., & Kahn, P. H. (2006). Value-sensitive design and information systems. In P. Z. Dennis Galletta (Ed.), *Human-computer interaction and management information systems: Applications* (pp. 348–372). New York: ME Sharpe.
- Gilbert, E., Karahalios, K., et al. (2008). The network in the garden. *Proceedings of CHI*. ACM
- Hayes, G. R. & Abowd G. (2006). Tensions in designing capture technologies for an evidence-based care community. *Proceedings of CHI* (pp. 937–946). Montreal, Quebec, Canada: ACM
- Hayes, G. R., Kientz, J. et al. (2004). Designing capture applications to support the education of children with autism. *Proceedings of UbiComp*. Tokyo, Japan: ACM
- Herring, S. (1996). LInguistic and critical analysis of computer-mediated communication: Some ethical and scholarly considerations. *The Information Society*, 12, 153–168.
- Hofmann, J. (1999). Writers, texts, and writing acts: Gendered user images in word processing software. In D. MacKenzie & J. Wajcman (Eds.), *The social shaping of technology*. New York: Open University Press.
- Hudson, J. M., & Bruckman, A. (2004). “Go Away”: Participant objections to being studied. *The Information Society*, 20(2), 127–139.
- IEEE (2012). *Software engineering code of ethics and professional practice*. Retrieved June 13, 2012 from http://www.computer.org/portal/web/certification/resources/code_of_ethics
- Kant, I. (1964). *Groundwork for the metaphysic of morals*. New York: Harper & Row.
- Kraut, R. (2012). *Personal communication to A. Munich, Germany*: Bruckmann.
- Kraut, R., Olson, J., et al. (2004). Psychological research online: Report of Board of Scientific Affairs’ Advisory Group on the Conduct of Research on the Internet. *American Psychologist*, 59(4), 1–13.
- LeDantec, C. A. & Edwards, W. K. (2008). Designs on dignity: Perceptions of technology among the homeless. *Proceedings of CHI* (pp. 627–636). Florence, Italy: ACM
- Mamykina, L. & Mynatt, E. (2007). *Proceedings of Healthnet* (pp. 49–54). San Juan, Puerto Rico: ACM
- Manier, J. (1999). *UIC suspended from doing most human research*. Chicago, IL, Tribune Company: Chicago Tribune.
- Nisbett, R. (2012). *Personal communication to A. Munich, Germany*: Bruckmann.
- OHRP (2001). *US Code of Federal Regulations Title 45 Part 46, Protection of Human Subjects*. United States Federal Code. Retrieved June 15, 2012 from <http://ohrp.osophs.dhhs.gov/humansubjects/guidance/45cfr46.htm>

- OHRP (2012). *Compliance oversight, determination letters*. From <http://www.hhs.gov/ohrp/compliance/letters/index.html>
- Rosaldo, R. (1993). *Culture and truth: The remaking of social analysis*. Boston, MA: Beacon.
- Seidman, I. (2006). *Interviewing as qualitative research*. New York: Teachers College Press.
- Silberman, M. S., Ross, J. et al. (2010). Sellers' problems in human computation markets. *Proceedings of KDD-HCOMP* (pp. 18–21). Washington, DC: ACM
- UNLV (2012). *History of research ethics*. Retrieved June 13, 2012 from <http://research.unlv.edu/ORI-HSR/history-ethics.html>
- Weiss, R. (1999). *US halts human research at duke*. Washington, DC: The Washington Post.